Practical No.: 3

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Problem Statement: Implement a circular queue using an array.

Perform following operations on it.

a. Insertion (Enqueue)

b. Deletion (Dequeue)

c. Display

Code:

```
#include <iostream>
using namespace std;

#define SIZE 5

class CircularQueue
{
  int queue[SIZE];
  int front, rear;

public:

  CircularQueue()
  {
    front = -1;
    rear = -1;
  }

  // Enqueue
```

```
void enqueue()
{
   int value;

   // Check Queue Overflow
   if (front == (rear + 1) % SIZE)
   {
      cout << "\nQueue Overflow";
      return;
   }

   cout << "\nEnter Element : ";
   cin >> value;

   if (front == -1)
   {
      front = rear = 0;
   }
   else
   {
      rear = (rear + 1) % SIZE;
   }

   queue[rear] = value;

   cout << "\nElement Inserted Successfully.";
}

// Dequeue
```

```
void dequeue()
{
   // Check Queue Underflow
   if (front == -1)
   {
      cout << "\nQueue Underflow";
      return;
   }

   cout << "\nDeleted Element : " << queue[front];

   if (front == rear)
   {
      front = rear = -1;
   }
   else
   {
      front = (front + 1) % SIZE;
   }
}

// Display
void display()
{
   if (front == -1)
   {
      cout << "\nQueue is Empty";
      return;
   }
```

```
        cout << "\nQueue Elements : ";

        int i = front;

        while (true)
        {
            cout << queue[i] << " ";

            if (i == rear)
                break;

            i = (i + 1) % SIZE;
        }

        cout << endl;
    }
};


int main()
{
    CircularQueue q;

    int choice;

    do
    {
        cout << "\n\n===== CIRCULAR QUEUE =====";
```

```
cout << "\n1. Enqueue";
cout << "\n2. Dequeue";
cout << "\n3. Display";
cout << "\n4. Exit";

cout << "\n\nEnter Choice : ";
cin >> choice;

switch (choice)
{
   case 1:
      q.enqueue();
      break;

   case 2:
      q.dequeue();
      break;

   case 3:
      q.display();
      break;

   case 4:
      cout << "\nProgram Terminated";
      break;

   default:
      cout << "\nInvalid Choice";
}
```

Output:

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter Choice : 1

Enter Element: 20

Element Inserted Successfully.

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter Choice : 1

Enter Element : 23

Element Inserted Successfully.

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter Choice : 2

Deleted Element : 20

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter Choice : 3

Queue Elements : 23

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter Choice : 4

Program Terminated

```
    } while (choice != 4);
```

```
    return 0;
}
```